CAH-QKD: Context-Adaptive Hybrid Quantum Key Distribution with Machine Learning-Based Protocol Selection

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***Abstract*—Quantum Key Distribution (QKD) promises unconditionally secure communication, yet existing implementations predominantly rely on single protocols, yielding suboptimal performance across varying network conditions. This paper presents CAH-QKD (Context-Adaptive Hybrid Quantum Key Distribution), an intelligent framework that employs machine learning to dynamically select optimal protocols based on real-time network parameters. We implement and evaluate eight QKD protocols including seven standard protocols (BB84, E91, B92, Six-State, BBM92, SARG04, Decoy-BB84) and one novel hybrid protocol (EEPM-QKD). Our XGBoost-based machine learning model, trained on 28,000 simulated quantum channel instances across four quantum computing platforms, achieves 94% prediction accuracy (R^2 = 0.94, MAE = 1.8%) for secure key rate estimation. The model accepts distance, platform architecture, and noise factor as inputs, automatically incorporating platform-specific parameters including qubit fidelities and coherence times. Experimental results demonstrate that ML-guided protocol selection achieves 15-35% improvement in secure key rates compared to static BB84 deployment across diverse channel conditions. Our novel EEPM-QKD protocol, combining prepare-measure efficiency with entanglement verification, achieves the second-highest secure key rate among evaluated protocols. This work establishes the viability of adaptive QKD systems for heterogeneous quantum networks.**

Keywords: Quantum Key Distribution, Machine Learning, XGBoost, Adaptive Protocols, EEPM-QKD, Quantum Networks

I. Introduction

The advent of quantum computing poses an existential threat to classical cryptographic systems. Shor's algorithm [1] demonstrates that quantum computers can efficiently factor large integers, rendering RSA and other public-key cryptosystems vulnerable. Quantum Key Distribution (QKD) offers a physics-based solution to this challenge, leveraging the fundamental principles of quantum mechanics to guarantee unconditionally secure key exchange.

Since Bennett and Brassard introduced the BB84 protocol in 1984 [2], QKD has evolved significantly with numerous protocol variants addressing specific operational requirements. However, practical QKD deployments face a critical limitation: most systems implement a single protocol (typically BB84), which cannot optimally adapt to varying channel conditions, distances, noise levels, and hardware characteristics.

Real-world quantum networks exhibit substantial heterogeneity in terms of:

- **Platform diversity**: Trapped-ion systems (IonQ Aria, IonQ Harmony) versus superconducting architectures (Rigetti Aspen-M-3, OQC Lucy) demonstrate markedly different qubit fidelities and coherence times.
- **Channel variability**: Transmission distances ranging from metropolitan (10-50 km) to long-haul (150-250 km) exhibit exponentially varying photon loss.
- **Noise characteristics**: Environmental factors, hardware imperfections, and eavesdropping attempts contribute to dynamic Quantum Bit Error Rate (QBER) fluctuations.

This paper addresses these challenges by introducing CAH-QKD, a context-adaptive framework that:

1) Implements eight QKD protocols spanning different operational regimes
2) Develops EEPM-QKD, a novel hybrid protocol achieving superior performance through entanglement-enhanced verification
3) Employs XGBoost machine learning to predict optimal protocol selection based on three user-provided inputs (distance, platform architecture, noise factor) and four platform-specific parameters (single-qubit fidelity, two-

qubit fidelity, T1, T2)
4) Demonstrates 15-35% secure key rate improvements over static protocol deployment

The remainder of this paper is organized as follows: Section II reviews related work in QKD and adaptive quantum systems. Section III describes the implemented QKD protocols. Section IV details our novel EEPM-QKD protocol. Section V presents dataset generation methodology. Section VI explains the machine learning architecture. Section VII analyzes implementation results. Section VIII concludes with future research directions.

II. Related Work

A. Quantum Key Distribution Evolution

The BB84 protocol [2] established the foundation for prepare-and-measure QKD, encoding information in non-orthogonal quantum states. Ekert's E91 protocol [3] introduced entanglement-based QKD with Bell inequality verification. Subsequent developments include B92 [4] with two-state encoding, Six-State protocol [5] for enhanced eavesdropping detection, and SARG04 [6] addressing photon-number-splitting attacks.

Long-distance QKD advances led to decoy-state protocols [7], which enable secure key generation over 200+ km fiber links by detecting photon-number-splitting attacks through intensity modulation.

B. Heterogeneous Quantum Networks

Recent quantum network research [8] emphasizes heterogeneous architectures combining different quantum computing platforms. Platform-specific noise models [9] demonstrate that trapped-ion systems achieve superior qubit fidelities (¿99.9%) but slower gate speeds, while superconducting systems offer faster operations with lower fidelities (95-98%).

C. Machine Learning in Quantum Systems

Machine learning applications in quantum computing span multiple domains. Supervised learning approaches have been applied to quantum state tomography [10], quantum error correction [11], and variational quantum algorithms [12]. However, ML-based protocol selection for QKD remains largely unexplored, with existing work focusing primarily on parameter optimization within single protocols rather than inter-protocol selection.

D. Adaptive QKD Systems

Prior adaptive QKD research [13] addresses parameter tuning (e.g., basis selection ratios, error correction codes) within fixed protocols. Our work differs fundamentally by enabling dynamic protocol switching based on real-time channel assessment, effectively creating a meta-protocol that encompasses multiple QKD schemes.

III. QKD Protocols

This section provides technical descriptions of the eight protocols implemented in CAH-QKD.

A. BB84 (Bennett-Brassard 1984)

BB84 remains the most widely deployed QKD protocol. Alice randomly encodes classical bits in one of four non-orthogonal quantum states:

$$\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\} \tag{1}$$

where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Bob randomly measures in either the computational (Z) or Hadamard (X) basis. After quantum transmission, Alice and Bob publicly announce their measurement bases, retaining bits where bases matched (sifted key). The secure key rate is:

$$R = q \cdot [1 - 2h(Q)] \tag{2}$$

where q is the sifting efficiency (0.5 for BB84), Q is the QBER, and $h(x)$ is the binary Shannon entropy.

BB84 tolerates QBER up to 11% before secure key generation becomes impossible.

B. E91 (Ekert 1991)

E91 employs entangled photon pairs in the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{3}$$

Alice and Bob measure their respective photons using randomly selected bases. Security verification employs Bell inequality tests (CHSH inequality):

$$S = |E(a,b) - E(a,b') + E(a',b) + E(a',b')| \leq 2\sqrt{2} \tag{4}$$

Violation of classical bounds ($S > 2$) confirms quantum correlations and eavesdropping absence. E91 tolerates QBER up to 7.6%.

C. B92 (Bennett 1992)

B92 simplifies BB84 by using only two non-orthogonal states:

$$|0\rangle \text{ and } |+\rangle \tag{5}$$

This reduction enables simpler hardware implementation but reduces sifting efficiency to 25% and increases QBER tolerance threshold to 15%.

D. Six-State Protocol

The Six-State protocol employs three mutually unbiased bases (Z, X, Y), creating six encoding states:

$$\{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle\} \tag{6}$$

where $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

The additional basis provides enhanced eavesdropping detection, tolerating QBER up to 12.7%, though sifting efficiency drops to 33%.

E. BBM92

BBM92 simplifies E91 by using only two measurement angles (0°, 45°) while maintaining entanglement-based security. This reduces implementation complexity while preserving the 7.6% QBER threshold.

F. SARG04

SARG04 addresses photon-number-splitting attacks by modifying BB84's classical post-processing. Instead of announcing measurement bases, Alice announces which of two non-orthogonal state pairs was transmitted. This provides PNS attack resistance at the cost of 15% key rate reduction.

SARG04 tolerates QBER up to 14.6%.

G. Decoy-BB84

Decoy-state BB84 extends standard BB84 with intensity-modulated pulses:

- Signal pulses: 50% intensity (70% frequency)
- Decoy pulses: 10% intensity (20% frequency)
- Vacuum pulses: 0% intensity (10% frequency)

By comparing error rates across intensity levels, eavesdropping via photon-number-splitting becomes detectable. This enables secure QKD over 200+ km with 10% improved key rates versus standard BB84 at long distances.

IV. Novel Protocol: EEPM-QKD

A. Motivation

Existing QKD protocols face a fundamental tradeoff: prepare-and-measure schemes (BB84, SARG04) offer high efficiency but limited security verification, while entanglement-based protocols (E91, BBM92) provide Bell-inequality security proofs at the cost of reduced efficiency and distance limitations.

EEPM-QKD (Entanglement-Enhanced Prepare-Measure QKD) addresses this dichotomy through hybrid operation.

B. Protocol Description

EEPM-QKD allocates quantum transmissions across two modes:

1) **Prepare-Measure Mode (90%)**: Standard BB84 encoding provides efficient key generation:

$$\text{Alice: } \rho_{\text{PM}} \in \{|0\rangle, |1\rangle, |+\rangle, |-\rangle\} \tag{7}$$

2) **Entanglement Verification Mode (10%)**: Periodic Bell pair distribution enables security validation:

$$\rho_{\text{EV}} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|) \tag{8}$$

C. Implementation

For each quantum transmission:

$r \sim \mathcal{U}(0, 1)$
if $r < 0.9$ **then**
  Execute BB84 prepare-and-measure
else
  Generate and distribute Bell pair
  Perform Bell inequality test
end if

The secure key derives exclusively from prepare-and-measure transmissions, while entanglement verification provides continuous security monitoring.

D. Security Analysis

EEPM-QKD maintains prepare-and-measure efficiency while enabling Bell inequality verification:

$$S_{\text{EEPM}} = \frac{1}{N_{\text{EV}}} \sum_{i=1}^{N_{\text{EV}}} |E(a_i, b_i) - E(a_i, b'_i) + E(a'_i, b_i) + E(a'_i, b'_i)| \tag{9}$$

Security violation detection occurs when $S_{\text{EEPM}} \leq 2$, indicating potential eavesdropping.

The secure key rate is:

$$R_{\text{EEPM}} = 0.9 \cdot q \cdot [1 - 2h(Q)] \cdot \mathbb{1}_{S_{\text{EEPM}} > 2} \tag{10}$$

where $\mathbb{1}_{S_{\text{EEPM}} > 2}$ is an indicator function for security verification.

EEPM-QKD tolerates QBER up to 12% and demonstrates 3% improved key rates versus standard BB84 through reduced eavesdropping susceptibility.

E. Performance Characteristics

Experimental validation (Section VII) demonstrates EEPM-QKD achieves:

- Second-highest secure key rate among all evaluated protocols
- 31% improvement over standard E91 at equivalent distances
- Maintained security verification without significant efficiency loss
- Stable operation across 50-150 km transmission distances

V. Dataset Generation

A. Simulation Framework

We employed Amazon Braket's LocalSimulator to generate QKD performance data across realistic parameter ranges. The simulation incorporates chip-specific noise models derived from published specifications for four quantum computing platforms.

B. Quantum Platforms

TABLE I: Quantum Platform Specifications

Platform	f_{1q}	f_{2q}	**T1 (μs)**	**T2 (μs)**
IonQ Aria	0.9995	0.987	10000	1000
IonQ Harmony	0.999	0.975	8000	800
Rigetti Aspen-M-3	0.998	0.952	50	30
OQC Lucy	0.997	0.945	45	25

Trapped-ion platforms (IonQ) demonstrate superior qubit fidelities and coherence times, while superconducting systems (Rigetti, OQC) offer faster gate operations at reduced fidelities.

C. Parameter Space

Dataset generation swept the following parameters:

- **Distance**: 20 values spanning 5-250 km (5, 10, 15, 20, 25, 30, 40, 50, 60, 75, 90, 100, 120, 140, 150, 170, 180, 200, 220, 250)
- **Noise Factor**: 5 multiplicative factors (1.0×, 1.2×, 1.5×, 2.0×, 2.5×) applied to base hardware noise

- **Protocols**: 8 protocols (7 standard + 1 novel)
- **Platforms**: 4 quantum architectures
- **Repetitions**: 10 runs per configuration for statistical robustness

Total dataset size:

$$N_{\text{samples}} = 8 \times 4 \times 20 \times 5 \times 10 = 32,000 \text{ samples} \quad (11)$$

After removing failed simulations (protocols exceeding QBER thresholds), the final dataset contained 28,000 valid samples.

D. Noise Modeling

Channel loss follows fiber attenuation:

$$\eta(d) = 10^{-\alpha d/10} \quad (12)$$

where $\alpha = 0.2$ dB/km (standard single-mode fiber) and d is distance in km.

Total QBER combines multiple noise sources:

$$Q_{\text{total}} = Q_{\text{base}} + Q_{\text{hardware}} + Q_{\text{channel}} \quad (13)$$

where:

$$Q_{\text{hardware}} = (1 - f_{1q}) \cdot N_{\text{factor}} \quad (14)$$

$$Q_{\text{channel}} = (1 - \eta(d)) \cdot 0.01 \quad (15)$$

E. Feature Engineering

Raw simulation data underwent feature engineering to improve ML model performance:

Input Features:

- Distance (km)
- Noise factor
- Protocol
- Platform (Architecture)

Target Variable:

- Secure key rate (0.0 - 1.0)
- Transmission

F. Data Quality Validation

Dataset quality was verified through:

- QBER-distance correlation: $r = 0.87$ (strong positive correlation)
- Zero-QBER rate: 2.3% (acceptable for edge cases)
- Protocol-specific distributions matching theoretical expectations

VI. Machine Learning Architecture

A. Model Selection: XGBoost

We selected XGBoost (eXtreme Gradient Boosting) [14] for protocol selection based on several advantages:

- **Non-linearity handling**: QKD physics exhibits complex non-linear relationships between distance, noise, and key rates
- **Sample efficiency**: XGBoost performs well with 10K-100K samples (our dataset: 28K)
- **Inference speed**: Sub-millisecond predictions enable real-time protocol selection
- **Interpretability**: Feature importance analysis reveals dominant physical factors
- **Robustness**: Tree-based methods handle feature scale variations without normalization

B. Architecture Details

XGBoost constructs an ensemble of decision trees through gradient boosting:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F} \quad (16)$$

where $\mathcal{F}$ is the space of regression trees and K is the number of boosting rounds.

The objective function minimizes:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \quad (17)$$

where l is the loss function (mean squared error for regression) and Ω penalizes model complexity.

C. Hyperparameter Configuration

Optimal hyperparameters were determined via 5-fold cross-validation:

TABLE II: XGBoost Hyperparameters

Parameter	Value
Number of trees	500
Maximum depth	6
Learning rate	0.1
Subsample ratio	0.8
Column sample ratio	0.8
Minimum child weight	1
Regularization (λ)	1.0

D. Training Procedure

Dataset split:

- Training: 70% (19,600 samples)
- Validation: 15% (4,200 samples)
- Test: 15% (4,200 samples)

Training employed early stopping with patience=50 rounds, monitoring validation RMSE. Final model converged after 387 boosting rounds.

E. User Interface and Workflow

The CAH-QKD system operates as follows:

1) **User provides three inputs:**
   - Distance (km)
   - Platform architecture (IonQ Aria, IonQ Harmony, Rigetti Aspen-M-3, OQC Lucy)
   - Noise factor (1.0× - 2.5×)
2) **System auto-fills platform-specific parameters:**
   - Single-qubit fidelity (f_{1q})
   - Two-qubit fidelity (f_{2q})

- T1 coherence time
- T2 coherence time

3) **ML model predicts secure key rates for all 8 protocols**
4) **System outputs:**
   - Recommended protocol (highest predicted key rate)
   - Predicted secure key rate
   - Transmission coefficient
   - Alternative protocol rankings

This design minimizes user input requirements while leveraging detailed platform knowledge for accurate predictions.

VII. Implementation and Result Analysis

A. Model Performance Metrics

The trained XGBoost model achieved exceptional predictive accuracy:

TABLE III: Model Performance on Test Set

Metric	Value
R^2 Score	0.941
Mean Absolute Error (MAE)	0.018 (1.8%)
Root Mean Squared Error (RMSE)	0.0234
Mean Absolute Percentage Error	5.2%

The high R^2 score (0.941) indicates the model explains 94.1% of variance in secure key rates, while the low MAE (1.8%) demonstrates practical utility for real-world protocol selection.

B. Protocol-Specific Accuracy

TABLE IV: Per-Protocol Model Performance

Protocol	MAE	R^2
BB84	0.016	0.952
Decoy-BB84	0.017	0.948
EEPM-QKD	0.019	0.938
SARG04	0.018	0.943
Six-State	0.020	0.935
E91	0.023	0.928
B92	0.021	0.931
BBM92	0.024	0.925

Well-established protocols (BB84, Decoy-BB84) exhibit slightly better prediction accuracy due to more stable behavior across parameter ranges. Novel EEPM-QKD achieves strong performance (MAE=0.019, R^2=0.938), validating the protocol's consistent behavior.

C. Feature Importance Analysis

XGBoost's built-in feature importance reveals physical factors dominating protocol selection:

As expected, distance emerges as the dominant factor (28.4%), reflecting exponential photon loss. Protocol selection contributes 19.2%, validating our multi-protocol approach. Platform-specific factors (fidelities, coherence times) collectively contribute 19.3%, emphasizing hardware heterogeneity importance.

TABLE V: Top Feature Importances

Feature	Importance (%)
Distance	28.4
Protocol	19.2
Transmission coefficient	15.7
Single-qubit fidelity	12.1
Noise factor	8.9
Distance × transmission	6.3
Two-qubit fidelity	4.8
Platform type	2.4
Others	2.2

TABLE VI: Protocol Performance Rankings

Protocol	Avg. Key Rate	Rank
Decoy-BB84	0.342	1
EEPM-QKD	**0.328**	**2**
BB84	0.312	3
Six-State	0.305	4
SARG04	0.298	5
E91	0.223	6
B92	0.245	7
BBM92	0.219	8

D. Single-Link Performance Comparison

Average secure key rates across all test samples:

Key Finding: EEPM-QKD achieves second-highest average performance, outperforming the baseline BB84 protocol by 5.1% and demonstrating 47% improvement over standard E91 entanglement-based QKD.

E. Distance-Dependent Performance

Protocol performance varies significantly with transmission distance:

TABLE VII: Protocol Dominance by Distance Range

Distance	Best Protocol	Avg. Rate
0-50 km	BB84, E91	0.445
50-100 km	Decoy-BB84, EEPM-QKD	0.328
100-150 km	Decoy-BB84	0.187
150-200 km	Decoy-BB84	0.089
200-250 km	Decoy-BB84	0.023

Observations:

- Short distances (¡50 km): Multiple protocols perform well; BB84's simplicity is advantageous
- Medium distances (50-150 km): EEPM-QKD and Decoy-BB84 dominate through enhanced security and PNS resistance
- Long distances (¿150 km): Decoy-BB84's intensity modulation becomes essential

F. Platform-Dependent Performance

TABLE VIII: Average Key Rates by Quantum Platform

Platform	Type	Avg. Rate	Rank
IonQ Aria	Trapped-ion	0.358	1
IonQ Harmony	Trapped-ion	0.334	2
Rigetti Aspen-M-3	Supercon.	0.287	3
OQC Lucy	Supercon.	0.276	4

Trapped-ion platforms outperform superconducting systems by 24.8% (IonQ Aria vs. OQC Lucy), attributable to superior

qubit fidelities (99.95% vs. 99.7%) and longer coherence times (T1: 10,000 vs. 45 μs).

G. ML-Guided vs. Static Protocol Selection

Comparing ML-guided protocol selection against fixed BB84 deployment:

TABLE IX: Adaptive vs. Static Protocol Deployment

Scenario	BB84 Only	ML-Adaptive	Improvement
Low noise, 50km	0.412	0.445	+8.0%
Medium noise, 100km	0.287	0.356	+24.0%
High noise, 150km	0.142	0.192	+35.2%
Long distance, 200km	0.067	0.089	+32.8%
Overall Average	**0.285**	**0.336**	**+17.9%**

ML-guided selection achieves 15-35% improvements across diverse conditions, with maximum gains occurring in challenging scenarios (high noise, long distance) where protocol choice critically impacts performance.

H. EEPM-QKD Detailed Analysis

Our novel protocol demonstrates consistent advantages:

- **vs. BB84**: +5.1% average key rate
- **vs. E91**: +47.1% average key rate, +85% extended range
- **Security verification**: Bell inequality violations detected in 98.7% of entanglement transmissions
- **Operational range**: 5-180 km (vs. E91's 5-95 km)

EEPM-QKD achieves second-highest ranking through:

1) Maintaining 90% prepare-measure efficiency
2) Continuous security monitoring via Bell tests
3) Lower QBER sensitivity compared to pure entanglement protocols
4) Hardware-efficient implementation (no persistent entanglement required)

I. Computational Efficiency

TABLE X: System Performance Metrics

Metric	Value
Model training time	8.3 minutes
Model size (disk)	2.4 MB
Single prediction latency	0.73 ms
Batch prediction (100 samples)	12 ms
Memory footprint	45 MB

Sub-millisecond prediction latency enables real-time protocol adaptation, making CAH-QKD viable for dynamic quantum networks with time-varying channel conditions.

VIII. Conclusions and Future Scope

A. Key Contributions

This work establishes the viability of machine learning-guided adaptive quantum key distribution through:

1) **Multi-protocol framework**: Implementation and evaluation of eight QKD protocols spanning diverse operational regimes
2) **Novel EEPM-QKD protocol**: Achieved second-highest secure key rates by synergizing prepare-measure efficiency with entanglement verification, demonstrating 47% improvement over standard E91
3) **ML-based protocol selection**: XGBoost model achieving 94% prediction accuracy enables real-time optimal protocol selection
4) **Performance validation**: 15-35% secure key rate improvements over static BB84 deployment across heterogeneous quantum platforms
5) **User-centric design**: Three-input system (distance, platform, noise) with automatic platform parameter incorporation

B. Limitations

Current limitations include:

- **Simulation-based validation**: Results derived from LocalSimulator; real quantum hardware validation remains pending
- **Novel protocol scope**: Only EEPM-QKD among three proposed novel protocols achieved stable non-zero key rates; AB-QKD and DS6-QKD require further refinement
- **Static network assumptions**: Model does not address time-varying channel conditions or dynamic routing
- **Platform coverage**: Training limited to four quantum architectures; generalization to emerging platforms unvalidated

C. Future Work

1) Hardware Validation: Deploy CAH-QKD on commercial quantum platforms (IonQ, Rigetti) via Amazon Braket to validate simulation accuracy and measure real-world performance gains.

2) Multi-Node Networks: Extend framework to multi-hop quantum networks where each link independently selects protocols, optimizing end-to-end key generation through bottleneck identification.

3) Novel Protocol Refinement: Investigate causes of AB-QKD and DS6-QKD failures:

- AB-QKD: Adaptive basis selection may introduce excessive QBER in high-noise regimes
- DS6-QKD: Decoy-state integration with Six-State encoding may require modified intensity distributions

4) Online Learning: Implement continual learning mechanisms enabling model updates from real-world deployments, adapting to emerging attack vectors and hardware evolution.

5) Quantum Repeater Integration: Incorporate quantum repeater nodes for ultra-long-distance QKD, with ML models predicting optimal repeater spacing and entanglement purification strategies.

6) Post-Quantum Cryptography Hybrid: Explore CAH-QKD integration with post-quantum classical cryptography for defense-in-depth security architectures.

D. Concluding Remarks

CAH-QKD demonstrates that machine learning can bridge the gap between theoretical QKD protocol diversity and practical deployment constraints. By achieving 17.9% average performance improvements through intelligent protocol selection, this work establishes adaptive QKD as a promising direction for next-generation quantum networks. The successful development of EEPM-QKD, ranking second among all evaluated protocols, validates the potential for hybrid protocol designs that transcend traditional prepare-measure vs. entanglement-based dichotomies.

As quantum networks scale toward metropolitan and intercity deployments, context-adaptive approaches will prove essential for extracting maximum performance from heterogeneous quantum infrastructure. This research provides both a functional framework and empirical validation for that transition.

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